

The Intersection of Artificial Intelligence and Higher Education

How the University of Minnesota can promote ethical uses of AI for
students and faculty alike.

Owen Peterson

July 8th, 2024

Executive Summary

Although artificial intelligence (AI) is still blooming, it is important that faculty and students adjust to this revolutionary and innovative technology. With fields, like healthcare, growing incredibly fast as a result of AI (Roumate, 2023, p. 138), it is vital that students acquire a solid foundation of understanding in the ways in which this technology can be used. It is also important that faculty implement this technology in their courses, which would allow students to get more experience with AI.

There are many mediums in which AI can be used for faculty, including instruction, course content, and administrative tasks. The field of AI has so much to offer instructors, ranging from preventing students from plagiarism to getting ideas for coursework to creating syllabus, quizzes, and exams. With this increased automation of tasks, instructors can, in turn, have more time to dedicate towards engaging with students.

As for students, there is an abundance of AI tools available to make learning more engaging and tailored to student's needs. Overcoming barriers for students, such as language and disability barriers, allow for easier access and a higher quality of learning. Getting help with course material is a significant use, as AI could explain information in a way that could be easier to understand in alternative ways and master. Despite these great benefits, there are shortcomings of AI that can weaken key skills of students in the future.

Keywords: artificial intelligence, knowledge objects, machine learning, learning analytics, data mining, meta-strategic knowledge, employability skills

Artificial intelligence (AI) - The development of machines that have some level of intelligence with the ability to perform human-like functions.

Knowledge objects - Tools used to discover and analyze aspects of your collected data, which can offer more ways to infer meaning from that data.

Machine learning - Knowledge discovery, the process of parsing based on sampling data sets known as training data, and generating meaningful patterns and a structured knowledge.

Learning analytics - The data collection of characteristics of students and knowledge objects in order to personalize educational methods to individual learner's needs and abilities.

Data mining - The process of collecting patterns from a large sample of data and developing inherent association rules.

Table of Contents

Executive Summary..... 1

Table of Contents.....2

Introduction.....3

Background..... 3

 Ethics in AI..... 3

 Key Elements Within Artificial Intelligence..... 4

Methodology..... 4

Results..... 4

Discussion..... 8

 Impacts on Academic Institutions, Faculty, and Students..... 9

Conclusion..... 10

Recommendations..... 10

References..... 12

Appendix A: “AI Usage in Academic Settings” Survey Questions..... 13

Introduction

The rapid development of artificial intelligence (AI) has prompted many questions about its future. Of these concerns, the intersection between AI and education is crucial to focus on, as it has the potential to impact student learning and development, as well as faculty's ability to perform administrative tasks more effectively and efficiently, and the way universities and other educational institutions adapt to ethical concerns of AI. This report aims to assess how AI impacts student learning and find how the University of Minnesota (UMN) can promote ethical usage of AI to mitigate academic dishonesty.

Background

While AI tools are still in their infancy, they have potential to alleviate learning and institutional management to a higher level. Former Microsoft CEO Bill Gates declared, "The development of artificial intelligence is as fundamental as the creation of the microprocessor, the personal computer, the Internet, and the mobile phone." As it stands, current AI systems do not have the ability to adapt to what the user needs, leading to learning feeling "one-size-fits-all" (Sharma et al., 2024, p. 4), but they are growing fast—according to the World Health Organization, "faster than international laws." Tools used to find AI-generated text, like zeroGPT and GPTZero, have been shown to be unreliable (Scott, 2024, p. 4). This sparks concern for the tools using AI, such as accuracy, reliability, and fairness. Due to this, a call for determining proper "ethics" of AI usage has been made by many agencies and even governments, including UNESCO, the US, and the European Commission.

Ethics in AI

The ways in which artificial intelligence can be used by companies, academic institutions, faculty, and students is concerning, as the data in these systems could be misused. Despite being used by about 68% of surveyed UMN students, AI is still in its infancy and is something that still needs to be addressed further in the coming years. The purpose of ethics in AI is to ensure transparent and explainable systems and "compliance between algorithm and fundamental rights" (Roumate, 2023, p. 135).

Organizations such as the Organisation for Economic Co-operation and Development have adopted their own recommendations on ethics in AI, consisting of five value-based principles, including human rights, well-being, democratic values, respect for the rule of law, and being bias free and easily accessible (Roumate, 2023, p. 138). The European Union has also adopted their own measures, including the single European data market by 2030, which would allow data to be stored in and retrieved from a single database. The report from a mandated AI governance order at the White House highlights transparency, accountability, and fairness as important principles to adopt in AI (Webber & Zheng, 2024, p. 6). To address the malicious use of AI in education and research, international standards and policy actions need to be discussed.

Key Elements Within Artificial Intelligence

Artificial intelligence has some key components that contribute to how it is used in a particular instance. These elements include, but are not limited to, machine learning, learning analytics, and data mining. Through machine learning, an AI program is able to discover new knowledge, parse based on sampling data sets known as training data, and then generate meaningful patterns and structured knowledge. Machine learning could be used, for instance, to grade student assignments, quizzes, and exams automatically. Learning analytics is the data of characteristics of students and knowledge objects in order to personalize educational methods to individual learner's needs and abilities. This would be used to tailor to individual student's needs and abilities, like providing instructional content and feedback on student work. Lastly, data mining is the process of collecting patterns from large samples of data and developing inherent association rules. An important application of data mining is personalized learning through the analysis of knowledge field data (Chen et al., 2020, p. 6).

Methodology

This study seeks to investigate the impact of AI on education, particularly on UMN students, as well as ways in which educational institutions such as UMN can implement and promote ethical AI usage. To assess this, a survey was conducted to learn more about how UMN students utilize AI in academic settings. The survey contained five qualitative questions, one quantitative question, and one optional open-ended question. The qualitative questions concerned the area of study of respondents, courses in which AI is used, frequency of use, ways of use, and AI tools being used for academic purposes, while the quantitative question addressed how helpful students find AI on a seven-point scale from 1 to 7, where 1 = not helpful at all, 4 = neutral, and 7 = extremely helpful. This numeric scale was chosen to gain a better idea of where students stand on the benefits of AI. The open-ended question was for any additional information that students had around AI usage in academics. This survey was shared in UMN spaces through social media platforms, such as Snapchat and Reddit, and responses were collected over the span of five days.

To explore more information about AI and its impact on education, research was conducted through Google Scholar, UMN library databases, such as Academic Search Premier and ACM, and Consensus, a database created via AI. Most of the sources are qualitative analyses focusing on the ways in which AI impacts education as a whole, including students, teachers, and institutions. This allowed for a more holistic look into AI's effect on education.

Results

The results of the survey are presented based on area of study, as well as the objectives and questions guiding this study. This survey showed how students implement artificial intelligence in their certain courses and fields of study.

Question 1:

Within the conducted survey, there were a total of 22 responses from UMN students. The first question collected the area of study of respondents, which was useful in getting a general understanding of the students using artificial intelligence the most in terms of major/minor studies. These demographics were divided into four groups. The first group covers **natural sciences**, which includes the likes of physics, biology, and chemistry; **formal sciences**, in the second group, containing majors such as mathematics, statistics, logic, and computer science; **professional and applied sciences**, which encompasses business, journalism, law, education, and engineering; and **humanities and social sciences**, which contains history, arts, and economics. Notably, the only group of students who all had reported usage of AI were formal science majors.

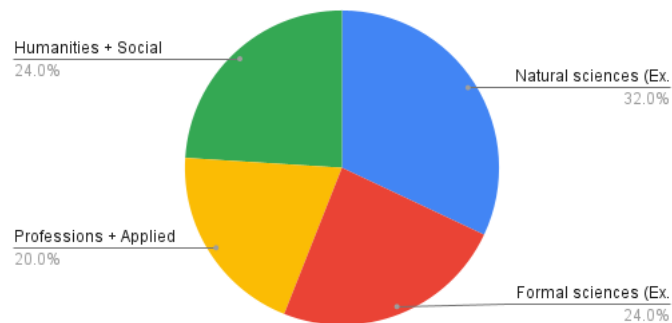


Figure 1: Area of study of respondents.

Question 2:

The second objective of the study was to measure how frequently artificial intelligence is being used among students. With there being a correlation between over-reliance on AI and students' weakened skills, including problem-solving and verbal communication, this is an essential question in finding how AI can be detrimental towards students' learning (Segbenya et al., 2023). As seen in figure 2, of the survey respondents, nearly 32% reported not using AI for educational purposes, and of the students who did report using AI, 60% use AI at most weekly.

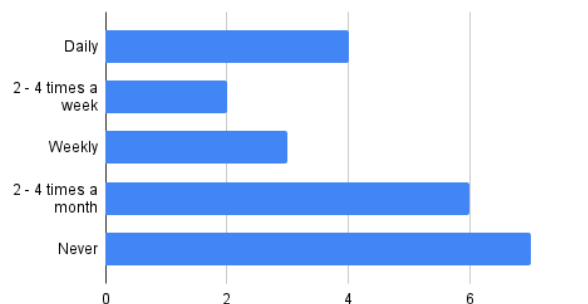


Figure 2: Frequency of AI use by respondents.

Question 3:

From the survey, data was found on the third objective of the survey, which looks at the courses students use artificial intelligence the most in. The two most common course fields in which AI is used academically are “humanities and social sciences” and “professions and applied sciences” which both account for 29% of students who signaled using AI. This data, alongside figure 1, shows that there is a significant tick up in AI usage for professions and applied sciences.

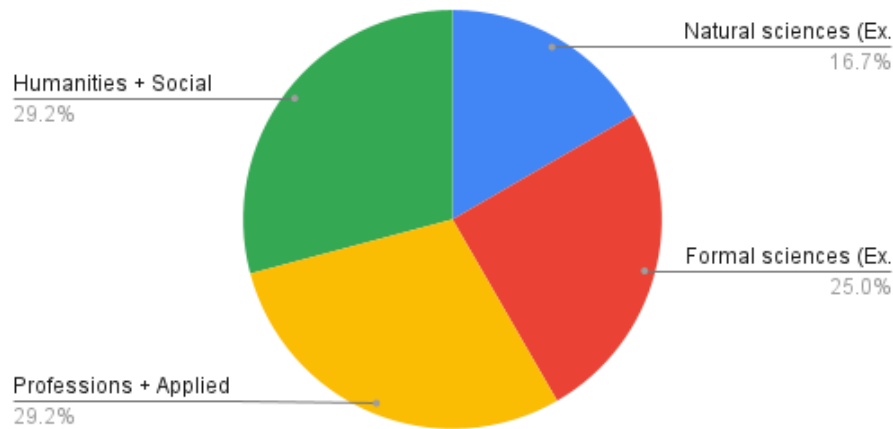


Figure 3: Courses most used with AI reported by students.

Question 4:

As for what artificial intelligence is used for in academic settings, the fourth question of the survey finds that of students who have reported AI use, about 33% use AI for comprehensive explanations on topics and 28% use it for further research. Regarding their coursework, only 8% said they used it for that purpose. Additionally, two students, accounting for 5%, reported that they use AI for feedback on writing and grammar checks.

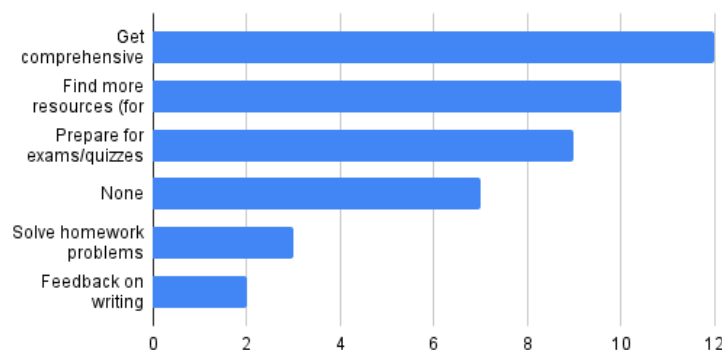


Figure 4: Ways in which AI is used in academic settings by respondents.

Question 5:

The fifth question of the survey was concerning the helpfulness of artificial intelligence in course material comprehension. The seven students who reported never using AI in academic ways scored it the lowest with an average of 1.14. The rest, who had used AI academically, put it significantly higher, with a mean of 5.6 and a median of 6. Of the whole survey, the mean was about 4.2, which remains relatively neutral.

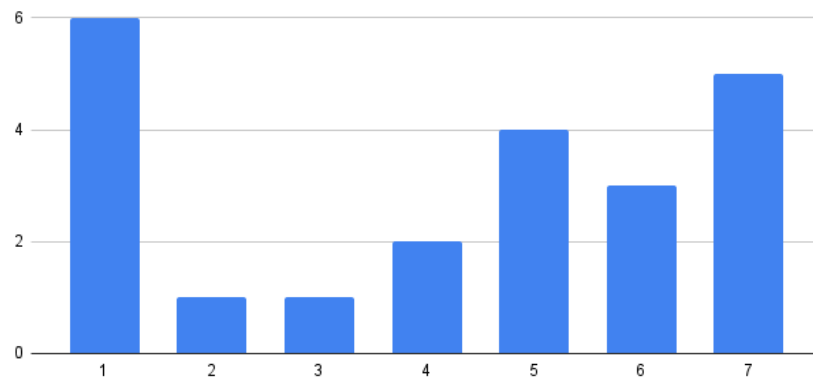


Figure 5: Students ranking of helpfulness of AI towards academics.

Question 6:

With artificial intelligence software becoming more widely used, this question addressed AI tools that students have found helpful. Of the options, ChatGPT easily wins as students' most popular choice, taking 67% of the poll. The next closest options were ClaudeAI and Quizlet AI which each had about 10% of student use. This highlights the fact that many uses of AI have not been made aware of to students.

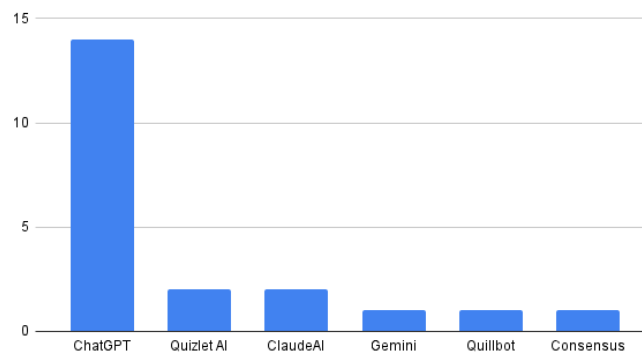


Figure 6: Helpful AI tools used by respondents.

Question 7:

The final question, which asked about any additional information on artificial intelligence usage, was only responded to by 6 people. One student said that the university's policy needs to be clarified, as some students are unaware of how they have used AI. This is something that needs to be defined and clarified for students, as without a proper understanding of the ways in which students are allowed to use AI and credit AI, students will unknowingly be engaging in academic dishonesty. Another student commented that the only consistently useful aspect of AI in academics is its discussion format, in that talking about a concept with AI is much more engaging than simply memorizing it; with more engagement, the memorization of information actually becomes easier for students. One respondent explained, "AI is like a textbook that can interact with students. Even if you have to verify the information, it can be extremely helpful for those who may not be able to make office hours."

Discussion

From the conducted survey, some interesting patterns were realized in the data. Of which, it is apparent that there is a direct relationship between frequency of use and measure of helpfulness of artificial intelligence. That is, as frequency of use increased, helpfulness measures increased. This is something that goes almost directly against the findings of Segbenya et al (2023, p. 6), which explains the inverse relationship between detrimental impacts of AI and AI usage—over-reliance on AI leads to weakened student development and skills. Additionally, the data collected shows that AI is significantly underused in the natural sciences field, as only 16% of respondents reported using AI in natural science courses, which is nearly half the percentage of respondents whose area of study is within natural sciences.

The findings from the survey are not without their concerns. Looking at the second question, a concern is sparked in that 18% of students reported daily use of AI. Interestingly, all of these same students gave AI 6's and 7's when rating its helpfulness. Despite that, is AI technology actually helping students with knowledge mastery and employability skills in the long term? The answer isn't as simple as it may seem, and will be discussed further down in this section. Another concern in the conducted survey is the question on AI usage in courses. The survey only accounted for a limited number of uses of AI, when there should have been more options for respondents. This caused an oversight when creating my survey in which "writing help", as well as other options such as "problem-solving" and "language translation", should have been considered in the creation of the survey.

Impacts on Academic Institutions, Faculty, and Students

Academic institutions

Following the COVID pandemic, most academic institutions, regardless of the level of education, had experienced ways in which they could use technology to engage with students. This has forced institutions to realize the various uses of technology, including e-learning, language translation, and increased automation, which all involve one way or another with artificial intelligence (Roumate, 2023, p. 131). Due to this new technology being implemented in much of today's technology, avoiding AI is no longer an option. One survey respondent expressed frustrations with the university, writing, "...the way UMN punishes students who are not aware that they have to report how they used AI is unfair and they need to be more clear within that policy." With that, clarification on AI usage policy and implementation of AI in the classroom are two principle steps that will be necessary for academic institutions in the coming future.

Faculty

Artificial intelligence provides a substantial amount of tools to course instructors and faculty. In the form of administrative tasks, AI would play a role in automatically reviewing, grading, and providing feedback on student's work, as well as proctoring students during exams.

Administrative tasks like creating syllabus, quizzes, and exams could even be done with AI (Atlas, 2023, p. 13). Through the use of technology, such as web-based platforms, robotics, and video conferencing, AI can improve upon instructional and learning quality and allow more time to be spent with students one-on-one (Chen et al., 2020, p. 12).

As for the content matter, instructors should integrate AI into course assignments in meaningful and relevant manners; "integrate AI into the tasks you are assigning in a way that mirrors how students might use it in the real world" (Scott, 2024, p. 8). Whether it be looking at AI generated prompts or checking work through AI tools, using this new technology should be promoted to give students a better understanding of the academic range in uses of AI. It is important that students learn about AI software and its limitations, as it will allow them to make informed decisions on what AI tools to use (Segbenya et al., 2023).

Students

Generally, there are great benefits to having artificial intelligence as a tool for students. These include global learning, overcoming language and disability barriers, and a more personalized learning experience. With more AI tools being developed every day, a qualified education is ensured for all students. The only concern is how developing countries will be able to access AI tools—due to AI being such a recent technology, it may not be as accessible as it could be for many students (Roumate, 2023, p. 131). The use of translation tools has caused overcoming language barriers to become even easier for students in the modern age (Atlas, 2023, p. 7).

Through data mining, academic content can be personalized and tailored to students educational needs, allowing for more engaging and adaptive learning experiences for students.

Within the classroom, AI tools can provide additional ways to ensure sustainable development. Some AI tools can be used to make scientific research easier, allowing for more information support for students (Roumate, 2023, p. 130). Students can also receive step-by-step help on problem-solving through complicated problems, allowing students to master knowledge and optimize learning more effectively (Chen et al., 2020, p. 10).

Despite all the benefits of AI, there are some concerns in regard to student usage. Of these concerns, academic dishonesty is of the utmost importance, as AI can generate text which can be used for writing assignments. This is pertinent to education, as students will be more inclined to use AI in order to complete assignments they do not see value in. Seeing as AI is still a developing technology, it has limitations, such as providing false or misleading responses to prompts and inventing citations, facts, and data to fill in the blanks (Scott, 2024, p. 4). With that, it is important that students practice not taking direct information or text from AI, as it can be subject to bias and inaccuracies, and making sure to find reputable sources that can backup AI-generated information. Additionally, the over-reliance on AI can be detrimental to students and their problem-solving, written and verbal communication, and employability skills (Segbenya et al., 2023). It is necessary for students to strike a balance in their usage of AI, as it can lead to alarming consequences on employability and job performance.

Conclusion

The purpose of this study was to learn more about how artificial intelligence impacts student learning and development and how AI usage can be pushed towards students in an academic setting. Through various measures, such as reconsidering current course curriculum and finding ways to integrate AI into that curriculum, students may not only engage more in course content, but also gain a better understanding of how they can use AI. At the same time, it is crucial that AI is not overused and relied upon—it should be another tool in a student's toolbox they can use to achieve academic success.

Overall, AI has been found to be a beneficial tool for students and has impacted them in a positive light. Though there are concerns on over-reliance and the detrimental effects on students, previous research has revealed a positive correlation between AI and the development of students, their abilities, and academic and employment success.

Recommendations

For future research on this fundamental topic, it is recommended that more uses of AI are accounted for when collecting data. Additionally, analyzing a larger, more diverse sample of

students in order to find patterns of use with AI in relation to fields of study. This could allow malicious uses to be more easily identifiable and more effectively prevented in the future.

Seeing how AI impacts other markets, like healthcare, it is fundamental that discussions be held on these topics which are extended to all sectors of the labor market. AI has the ability to replace jobs and in turn, shift external markets, and due to these implications, future research should not be limited to just computer science, or even STEM. Professionals working in the field of AI, as well as education, including professors and other faculty, should be engaged with regards to the issue of and what their experiences have been with AI in academics, its impact on students, and any policy making recommendations they may have.

References

- Atlas, Stephen. "ChatGPT for Higher Education and Professional Development: A Guide to Conversational AI." (2023). https://digitalcommons.uri.edu/cba_facpubs/548.
- Chen, L., Chen, P., & Lin, Z. (2020). Artificial Intelligence in Education: A Review. *IEEE Access*, 8, 75264-75278. <https://doi.org/10.1109/ACCESS.2020.2988510>.
- Roumate, F. (Ed.). (2023). *Artificial intelligence in higher education and scientific research : future development*. Springer. <https://doi.org/10.1007/978-981-19-8641-3>.
- Scott, I. (2024). Rising to Meet the Challenge of Generative AI. *Journal of Legal Studies Education*, 41(1), 29–37. <https://doi-org.ezp1.lib.umn.edu/10.1111/jlse.12141>.
- Segbenya, M., Bervell, B., Frimpong-Manso, E., Otoo, I. C., Andzie, T. A., & Achina, S. (2023). Artificial intelligence in higher education: Modelling the antecedents of artificial intelligence usage and effects on 21st century employability skills among postgraduate students in Ghana. *Computers and Education. Artificial Intelligence*, 5, 100188-. <https://doi.org/10.1016/j.caeai.2023.100188>.
- Sharma, S., Raj, M., Ali, M. M., Sharma, K., Singh, A., & Singh, S. (2024). Design of Structure using an Adaptive Generative AI System for Customised Education for Individuals. *2024 IEEE International Conference on Computing, Power and Communication Technologies (IC2PCT)*, 5, 276–280. <https://doi.org/10.1109/IC2PCT60090.2024.10486396>.
- Webber, K. L., & Zheng, H. Y. (2024). Artificial intelligence and advanced data analytics: Implications for higher education. *New Directions for Higher Education*, 1. <https://doi-org.ezp2.lib.umn.edu/10.1002/he.20508>.

Appendix A: “AI Usage in Academic Settings” Survey Questions

1. What is your area of study? (Major/minor)
 - a. Humanities + Social sciences (Ex. History, Arts, Econ)
 - b. Natural sciences (Ex. Physics, Biology, Chemistry)
 - c. Formal sciences (Ex. Mathematics, Statistics, Logic, CS)
 - d. Professions + Applied sciences (Ex. Business, Journalism, Law, Education)
 - e. Undeclared
 - f. Other...

2. Estimate how frequently you are using AI during the semester:
 - a. Never
 - b. 2-4 days a month
 - c. Weekly
 - d. 2-4 days a week
 - e. Daily

3. Which courses do you use AI the most for?
 - a. Humanities + Social sciences (Ex. History, Arts, Econ)
 - b. Natural sciences (Ex. Physics, Biology, Chemistry)
 - c. Formal sciences (Ex. Mathematics, Statistics, Logic, CS)
 - d. Professions + Applied sciences (Ex. Business, Journalism, Law, Education)
 - e. None
 - f. Other...

4. In what ways have you used AI in your courses?
 - a. Get comprehensive explanations on course topics
 - b. Solve homework problems
 - c. Find more resources (for research or general understanding)
 - d. Exam/quiz preparation
 - e. None
 - f. Other...

5. How helpful is AI towards your engagement and understanding of course material?
 - 1) Not helpful at all
 - 2)
 - 3)
 - 4)
 - 5)
 - 6)
 - 7) Extremely helpful

6. Which AI programs are you using?
 - a. ChatGPT
 - b. Quizlet AI
 - c. Consensus
 - d. Quillbot
 - e. Other...

7. Is there any other additional information on AI usage at UMN and how it has impacted you academically that you would feel comfortable sharing?